Part A

```
mininet@mininet-vm:~$ chmod +x LinearTopo.py
mininet@mininet-vm:~$ sudo mn --custom LinearTopo.py --topo mytopo
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s3 s4
*** Adding links:
(h1, s3) (s3, s4) (s4, h2)
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 2 switches
s3 s4 ...
*** Starting CLI:
mininet>
```

```
mininet> iperf
*** Iperf: testing TCP bandwidth between h1 and h2
*** Results: ['31.5 Gbits/sec', '31.6 Gbits/sec']
mininet>
```

```
mininet> exit
*** Stopping 1 controllers
c0
*** Stopping 3 links
...
*** Stopping 2 switches
s3 s4
*** Stopping 2 hosts
h1 h2
*** Done
completed in 463.632 seconds
mininet@mininet-vm:~$
```

Part B:

```
mininet@mininet-vm:~$ chmod +x LinearTopo
LinearTopoConstr.py  LinearTopo.py        LinearTopo.py.1
mininet@mininet-vm:~$ chmod +x LinearTopoConstr.py
mininet@mininet-vm:~$ sudo mn --custom LinearTopoConstr.py --topo mytopo --link
tc
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1 s2
*** Adding links:
(h1, s1) (10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (s1, s2) (s2, h2)
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 2 switches
s1 s2 ...(10.00Mbit 5ms delay) (10.00Mbit 5ms delay)
*** Starting CLI:
mininet> iperf
*** Iperf: testing TCP bandwidth between h1 and h2
*** Results: ['9.57 Mbits/sec', '10.8 Mbits/sec']
mininet>
```

Part C:

```
mininet@mininet-vm:~$ sudo mn --custom LinearTopo.py --topo mytopo --link tc
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4
*** Adding switches:
s3 s4
*** Adding links:
(h1, s3) (h2, s3) (100.00Mbit 10ms delay) (100.00Mbit 10ms delay) (s3, s4) (s4, h3) (s4, h4)
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller
c0
*** Starting 2 switches
s3 s4 ...(100.00Mbit 10ms delay) (100.00Mbit 10ms delay)
*** Starting CLI:
mininet> perf
*** Unknown command: perf
mininet> iperf
*** Iperf: testing TCP bandwidth between h1 and h4
*** Results: ['91.8 Mbits/sec', '93.3 Mbits/sec']
mininet>
```